Symptom'. Given that these relations can be expressed in a high level language such as English in myriad different ways, any information extraction system needs to be able to analyse the language syntactic constructs, and if needed, infer semantic meaning and extract the relations.

The structure used for representing the information extracted from the unstructured text is known formally as 'ontology'. It is the formal naming of the various entities of interest in a domain along with their associated relationships. There are two types of information extraction or IE systems—closed IE and open IE. The former is associated with a fixed set of entities and relations already specified through an ontology. In our example, if the ontology is pre-specified and is used as the basis for extracting information from the unstructured text, then it is closed IE, since only relations specified in the ontology and entities belonging to the types/classes of the ontology would be extracted from the text. Closed IE systems are typically used in domain-specific IE, where the entity classes and relationships are well defined and known. Some well-known examples of closed IE systems are NELL and DBpedia. NELL is the Never Ending Language Learning System *(http://rtw.ml.cmu.edu/rtw/)*, which uses machine learning techniques to continuously extract facts from the Web and represent it in a structured ontology. DBpedia is a structured representation of information extracted from unstructured Wikipedia texts *(http://wiki.dbpedia.org/)*. Both of these popular systems are closed IE systems since they extract only predefined relationships from unstructured texts.

On the other hand, open IE systems extract any relation that is present in unstructured text. Open IE systems do not need a pre-specified ontology to guide the extraction process. In our example above of extracting medical relations, consider a sentence of the form: 'Drug X has adverse side effects such as severe allergic cough and skin rash.' Since closed IE systems extract only the relations specified in the ontology, the side-effect relation will not be extracted by the closed IE system. However, this is an important fact associated with Drug X and can be extracted only by open information systems. Open IE systems do not have fixed ontology or entity types. All the possible relations present in the unstructured text are extracted irrespective of the type of entities and relationships.

After extracting all possible relations, an ontology may be constructed to populate the relations extracted in an organised structure. In certain cases, where a predetermined ontology is provided, the ontology can be extended if the open IE systems have extracted newer entity types or newer relation types. Open IE systems are typically used in Web-scale information systems by search engines. Well known examples of open IE systems include the Open IE 4.0 project *(https://github.com/knowitall/openie)*, the TextRunner project *(http://openie.allenai.org/)* and the Stanford openIE *(http://nlp.stanford.edu/software/openie.html)*.

OpenIE systems typically extract relation triplets of the form <*subject, predicate, object*> from unstructured text. For example, consider the following sentence from a text document: "Obama, president of the United States was born in Hawaii". OpenIE systems extract two relations of the form: <*"Obama", "is (being)", "president of United States"*> and <*"Obama", "born in", "Hawaii"*>. In both these relations, "Obama" is the subject. "born in" and "is (to be)" are the predicates in the two relations. "President of the United States" and "Hawaii" are the subjects in the two relations. While closed IE systems would need to be specified, OpenIE systems can discover the exact type of relations such as 'born in' automatically due to the language structure.

Relation extractors for closed IE systems need to be provided with relation structures and entity types, whereas open IE systems do not need this information. This allows open IE systems to discover much larger sets of relations than closed IE systems. For example, closed IE systems such as NELL and DBpedia support only 500 and 940 relation types, respectively. On the other hand, open IE systems such as TextRunner can extract 100,000+ relations, and ReVerb *(http://reverb.cs.washington.edu/)* can extract 5,000,000+ relations.

Extracting relations from unstructured text can be done by means of: (a) supervised, (b) weakly supervised, and (c) distant supervised techniques. The supervised approach for relation extraction requires extensive training examples of extractions. On the other hand, the weakly supervised approach only requires a handful of seed patterns, which can guide the extractor to iteratively identify further relation patterns. While weakly supervised techniques do not require extensive labelled data, they still need to be provided seed patterns for relation extraction. Distantly supervised techniques use the relationship facts available in external knowledge bases like DBpedia or Wikipedia to identify the initial seed patterns automatically, without the need for any human intervention to provide to the relation extractor systems. We will discuss each of these approaches in the next column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, happy programming! END

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